

Constant Magnetic Flow on the Hyperbolic Plane: Orbit Trichotomy, Critical Horocycles, and Exact Periods

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Abstract

On the complete simply connected surface of curvature $-\kappa^2$ we solve $D_t\dot{\gamma} = bJ\dot{\gamma}$ at fixed speed v . For $v > 0$, every trajectory is a circle, horocycle, hypercycle, or geodesic, according as $|b|$ is greater than, equal to, or less than κv (with $b = 0$ separated). Only the circle chamber closes, with primitive period $2\pi/\sqrt{b^2 - \kappa^2 v^2}$. At equality the Lorentz-frame generator is nonzero nilpotent, and the orbit is a nonclosed horocycle. We prove the classification globally from a raw 3×3 moving-frame equation. An independent radius–circumference calculation checks the period and fixes all orientation and normalization conventions.

1 Model, conventions, and result

Let $\mathbb{H}_\kappa^2$ be the oriented complete simply connected Riemannian surface with Gaussian curvature $-\kappa^2$, where $\kappa > 0$, and let J denote positive rotation through $\pi/2$. We study

$$D_t\dot{\gamma} = bJ\dot{\gamma}, \quad |\dot{\gamma}| = v, \quad (1)$$

for constant $b \in \mathbb{R}$. Metric compatibility makes the speed constant; we nevertheless retain v because replacing b by the geodesic curvature is a frequent source of a missing factor. Time in every period statement is the physical parameter t in (1).

The orbit geometry is classical. Comtet treats the hyperbolic constant-field problem and its Landau quantization [1]; Adachi treats magnetic trajectories on constant-holomorphic-curvature spaces [2]. No literature originality is claimed for the classification below. Our purpose is a convention-complete derivation joined to independently replayable evidence.

Theorem 1 (Complete orbit atlas). *Assume $\kappa > 0$ and $v > 0$. Every solution of (1) is complete. Up to an orientation-preserving isometry of $\mathbb{H}_\kappa^2$ and a time translation:*

1. *if $|b| > \kappa v$, it is a hyperbolic circle and its primitive period is*

$$T = \frac{2\pi}{\sqrt{b^2 - \kappa^2 v^2}}; \quad (2)$$

2. *if $|b| = \kappa v$, it is a nonclosed horocycle;*
3. *if $0 < |b| < \kappa v$, it is an unbounded hypercycle;*
4. *if $b = 0$, it is a geodesic.*

The sign of b reverses the orientation but not the unoriented curve or its period. At the critical face the ambient generator has $A^3 = 0$ and $A^2 \neq 0$.

Boundary convention. If $v = 0$, the curve is stationary and the unit-tangent proof below is not invoked. The flat case $\kappa = 0$ is a limit, not a point in the theorem: $b \neq 0$ gives a Euclidean circle of period $2\pi/|b|$, whereas $b = 0$ gives a straight line.

2 The constant Frenet system

For $v > 0$ put $T = \dot{\gamma}/v$ and $N = JT$. Equation (1) becomes

$$D_t T = bN, \quad D_t N = -bT, \quad (3)$$

so the signed geodesic curvature with respect to arclength is $k_g = b/v$. Use the hyperboloid

$$\mathbb{H}_\kappa^2 = \{X \in \mathbb{R}^{2,1} : \langle X, X \rangle_L = -\kappa^{-2}, X_0 > 0\}, \quad \eta = \text{diag}(-1, 1, 1).$$

Set $e_0 = \kappa X$. The ambient derivative of a tangent field Y along X is

$$\frac{dY}{dt} = D_t Y + \kappa^2 \langle \dot{X}, Y \rangle X, \quad (4)$$

or, in particular, $\dot{T} = \kappa v e_0 + bN$. With the signs in (3), let $F = (e_0, T, N)$ explicitly denote the matrix whose columns are those three vectors. It satisfies the right-action equation $\dot{F} = FA$, with

$$A = \begin{pmatrix} 0 & \kappa v & 0 \\ \kappa v & 0 & -b \\ 0 & b & 0 \end{pmatrix}, \quad A^\top \eta + \eta A = 0. \quad (5)$$

Changing from row to column frames transposes the displayed ODE but changes neither the characteristic polynomial nor any conclusion.

Lemma 1 (Cubic identity). *For (5),*

$$\det(\lambda I - A) = \lambda(\lambda^2 + b^2 - \kappa^2 v^2), \quad A^3 = (\kappa^2 v^2 - b^2)A. \quad (6)$$

At $|b| = \kappa v$, $A^2 \neq 0$ and $A^3 = 0$.

Proof. Direct expansion of the three-by-three determinant proves the first identity; Cayley–Hamilton proves the second. Substitution $b = \pm \kappa v$ leaves, for example, the $(1, 1)$ entry of A^2 equal to $\kappa^2 v^2 > 0$, so the critical generator is nonzero of nilpotency index three. $\square$

3 Proof of the classification

Proof of Theorem 1. The frame equation has the global solution $F(t) = F(0)e^{tA}$. Thus every initial oriented frame gives a solution for all t , and transitivity of $\text{SO}^+(2, 1)$ on such frames shows that the analysis exhausts all magnetic initial data.

If $\delta = b^2 - \kappa^2 v^2 > 0$, (6) gives eigenvalues $0, \pm i\sqrt{\delta}$. The one-parameter Lorentz subgroup is elliptic and fixes a timelike axis. Intersecting the hyperboloid with the affine plane at fixed Lorentz distance from that axis gives a hyperbolic circle. On the two-dimensional rotating block, put $e_0 = (1, 0, 0)^\top$ in frame coordinates. The cubic identity gives

$$e^{tA} e_0 = e_0 + \frac{\sin \theta}{\sqrt{\delta}} A e_0 + \frac{1 - \cos \theta}{\delta} A^2 e_0, \quad \theta = t\sqrt{\delta}.$$

Here $(Ae_0)_T = \kappa v \neq 0$, $(A^2 e_0)_T = 0$, and $(A^2 e_0)_{e_0} = \kappa^2 v^2 \neq 0$. A position return first forces $\sin \theta = 0$ from the tangent component and then forces $\cos \theta = 1$ from the e_0 component. Thus the base point returns exactly for $\theta \in 2\pi\mathbb{Z}$, and (2) is primitive.

If $\delta = 0$, Lemma 1 gives a nontrivial parabolic subgroup. Its hyperboloid orbits are horocycles. More specifically, $e^{tA} e_0 = e_0 + tAe_0 + t^2 A^2 e_0/2$, whose tangent component is κvt . It is nonzero for every $t \neq 0$, so the base point itself never returns; the critical horocycle is not closed.

If $\delta < 0$, the nonzero eigenvalues are real and the subgroup is hyperbolic. Its nongeodesic hyperboloid orbits are hypercycles. When $b = 0$, the normal is parallel and the curve is the invariant geodesic itself. The sign change $b \mapsto -b$ is conjugated by reversing N , so it reverses orientation and preserves the remaining data. $\square$

4 Independent geometric recovery of the period

The matrix proof already establishes the theorem. A second derivation checks that the clock and curvature conventions agree. A circle of hyperbolic radius ρ on curvature $-\kappa^2$ has unsigned geodesic curvature and circumference

$$|k_g| = \kappa \coth(\kappa\rho), \quad L(\rho) = \frac{2\pi}{\kappa} \sinh(\kappa\rho). \quad (7)$$

Because $|k_g| = |b|/v$, a finite ρ exists exactly when $|b| > \kappa v$. Moreover

$$\sinh^2(\kappa\rho) = \frac{1}{\coth^2(\kappa\rho) - 1} = \frac{\kappa^2 v^2}{b^2 - \kappa^2 v^2}.$$

Dividing $L(\rho)$ by v reproduces (2). Since this is an embedded circle traversed with nonzero constant speed and fixed orientation, one circumference is its first position return, independently confirming primitivity. As $|b| \downarrow \kappa v$, the radius and the period diverge and the circles converge, after recentering, to a horocycle. If $0 < |b| < \kappa v$, a curve at signed distance r from a geodesic has $|k_g| = \kappa |\tanh(\kappa r)|$; this gives the hypercycle relation

$$|\tanh(\kappa r)| = \frac{|b|}{\kappa v} < 1. \quad (8)$$

Thus the geometric and Lorentz classifications agree at every open chamber and at their common boundary.

5 What completeness does and does not assert

Global existence here follows from the constant frame exponential, rather than from a compactness shortcut (the hyperbolic plane is noncompact). The classification is of unparametrized shapes together with their physical-time parametrization. It does not identify different field strengths after a time change. Nor does elliptic subgroup type assert dynamical stability of a perturbed nonconstant-field system; no such perturbation theorem is made.

The critical nilpotent statement is deliberately stronger than observing a triple zero characteristic polynomial: the latter alone would not exclude $A = 0$ or a lower Jordan index. The explicit check $A^2 \neq 0$ fixes the parabolic orbit and its polynomial frame growth.

References

- [1] A. Comtet, *On the Landau Levels on the Hyperbolic Plane*, Annals of Physics 173 (1987), 185–209. [doi:10.1016/0003-4916\(87\)90098-4](https://doi.org/10.1016/0003-4916(87)90098-4).
- [2] T. Adachi, *Kaehler Magnetic Flows for a Manifold of Constant Holomorphic Sectional Curvature*, Tokyo Journal of Mathematics 18 (1995), 473–483. [doi:10.3836/tjm/1270043477](https://doi.org/10.3836/tjm/1270043477).